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Overcoming limitations of modelling rare species by using ensembles of small models

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Summary

1. Species distribution models (SDMs) have become a standard tool in ecology and applied conservation biology. Modelling rare and threatened species is particularly important for conservation purposes. However, modelling rare species is difficult because the combination of few occurrences and many predictor variables easily leads to model overfitting. A new strategy using ensembles of small models was recently developed in an attempt to overcome this limitation of rare species modelling and has been tested successfully for only a single species so far. Here, we aim to test the approach more comprehensively on a large number of species including a transferability assessment.

2. For each species, numerous small (here bivariate) models were calibrated, evaluated and averaged to an ensemble weighted by AUC scores. These 'ensembles of small models' (ESMs) were compared to standard SDMs using three commonly used modelling techniques (GLM, GBM and Maxent) and their ensemble prediction. We tested 107 rare and under-sampled plant species of conservation concern in Switzerland.

3. We show that ESMs performed significantly better than standard SDMs. The rarer the species, the more pronounced the effects were. ESMs were also superior to standard SDMs and their ensemble when they were evaluated using a transferability assessment.

4. By averaging simple small models to an ensemble, ESMs avoid overfitting without losing explanatory power through reducing the number of predictor variables. They further improve the reliability of species distribution models, especially for rare species, and thus help to overcome limitations of modelling rare species.

Key-words: AUC, BIOMOD, bivariate models, Boyce index, consensus forecast, endangered species, ensemble prediction, species distribution modelling, Switzerland

Introduction

Species distribution models (SDMs) are powerful tools for analysing environmental drivers of species distributions and for projecting a species' realized niche into a geographic area. The number of peer-reviewed papers using SDMs has increased strongly in recent years (Guisan *et al.* 2013) and applications of SDMs to conservation are manifold (Guisan & Thuiller 2005; Franklin 2009; Peterson *et al.* 2011). Such applications are especially important for rare and threatened species with high conservation priority (Guisan *et al.* 2013).

Models of rare species, characterized by narrow ecological ranges, often show higher accuracy compared to common species characterized by broad ecological ranges (Franklin *et al.* 2009 and literature mentioned there). However, model accuracy of rare species increases asymptotically with sample size (Stockwell & Peterson 2002; Hernandez *et al.* 2006) because the more the occurrences are available, the more complete the species' niche is described. Accordingly, many studies show increasing model accuracy with number of occurrences (e.g. Hernandez *et al.* 2006; Guisan *et al.* 2007; Wisz *et al.* 2008; see also references mentioned there) and below about 30 occurrences model accuracy is often low and variability of model accuracy increases across species (Stockwell & Peterson 2002; Hernandez *et al.* 2006; Wisz *et al.* 2008).

A major problem with modelling rare species is that there is frequently a high number of explanatory environmental variables in relation to few occurrences, which can lead to model overfitting, that is the fit between predicted values and actual data in models with a large number of predictors is misleadingly good. Overfitting leads to reduced generalizability of the model and thus restricts the applicability of the model to new data (Vaughan & Ormerod 2005).

This problem could be alleviated by reducing the number of predictors in a model. A general rule of thumb is that sample size (number of occurrences, i.e. in this context number of presences) should be 10 times larger than the number of predictors used for modelling (Harrell, Lee & Mark 1996). The number

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of predictors can be reduced by stepwise variable selection methods (Parviainen *et al.* 2008), for example based on information criteria like the Akaike Information Criterion (AIC, Akaike 1974) and the Bayesian information criterion (BIC, Schwarz 1978), by shrinkage methods such as the Lasso (Tibshirani 1996) or using cross-validation for variable selection. However, for rare species, modelling is often difficult because keeping a 1:10 ratio would mean to include only two predictors for 20 occurrences, which is usually insufficient information of a species' niche for modelling its distribution accurately.

The mismatch between the need to model rare species distributions for conservation purposes and the inherent difficulty in modelling them has been termed in the study of Lomba *et al.* (2010) the 'rare-species modelling paradox'. They proposed a new strategy for modelling the distribution of rare species based on fitting a larger number of small bivariate models, that is models with only two predictors at a time (although only one or three could also be used), and averaging them in an ensemble prediction using weights based on model performances. The predictor-to-occurrence ratio is improved greatly with this method because the number of predictors used in each small model remains low. The estimation of model coefficients was thus expected to be much more robust and the risk of model overfitting to be minimized without losing explanatory power because the number of predictors is kept low for a single model but high within the ensemble framework.

These ensembles of small models (ESMs) have been successfully applied so far only once for a single, rare and endemic plant species from the Iberian Peninsula (Lomba *et al.* 2010). The authors found an excellent ESM with an AUC score of 0·99. However, ESMs have not been compared so far to standard SDMs. Because ESMs are thought to be particularly useful when applied to rare species, the question is how rare a species must be so that ESMs are outperforming standard approaches (if at all). Lomba *et al.* (2010) evaluated their ESMs by conventional repeated split sampling only. It is thus open if ESMs would also outperform standard methods when transferred to new areas. In this study, we tested the hypotheses that (i) the performance of ESMs compared to commonly used standard ensembled SDMs is advantageous for very rare species where overfitting is a problem and that (ii) ESMs lose their advantage as the number of occurrences increases.

We focused on a set of 107 rare to very rare plant species of conservation concern in Switzerland. We investigated whether and when ESMs outperform standard ensembles of SDMs and included a transferability assessment to test the generality of ESMs through their ability to predict to new areas.

Methods

We fitted both ESMs and standard ensembles of SDMs with three widely used modelling techniques: generalized boosted models (GBMs), generalized linear models (GLMs) and maximum entropy (Maxent). We used GLMs because of their user-friendly and common usage, and GBMs (also termed boosted regression trees, BRTs) as well as Maxent models because of their good performance (Elith *et al.* 2006; Elith & Graham 2009; Heikkinen, Marmion & Luoto 2012).

ENSEMBLE OF SMALL MODELS

The basic idea behind ESMs is to model a species distribution based on small, simple models, for example all possible bivariate models (i.e. models that contain only two predictors at a time out of a larger set of predictors), and then combine all possible bivariate models into an ensemble (Lomba *et al.* 2010). In our study, a set of 11 predictors (Table 1) resulted in 55 bivariate predictor combinations. These steps are described in detail below. We evaluated each of the bivariate models by the cross-validated AUC index, which was then used to build a Somers' D weighted average of the 55 bivariate models as the ESM prediction (Fig. 1). Somers' D (also known as Gini coefficient) is $D = 2 \times (AUC - 0{\cdot}5)$ and gives more weight to models that perform well and less to those that perform poorly. Bivariate models with a Somers' D lower than 0 (i.e. $AUC < 0{\cdot}5$) were set to zero and not used to build the ESMs. Three ESMs were built separately, one for each of the three modelling techniques (i.e. ESMs based on bivariate GLM, bivariate GBM and bivariate Maxent model). We then built a final ensemble prediction by averaging across these three ESMs, again using Somers' D weights (ESM_{EP}; Fig. 1).

To get further insights into how ESMs are working, we assessed on three species how the performance of ESMs is changing when increasing the number of bivariate models starting with only one up to 55.

TARGET SPECIES

We selected 107 rare or under-sampled vascular plant species in Switzerland. Species are considered under-sampled if only low numbers of occurrences are available although the species are known to be more common (in the following, we group under-sampled and rare species together and refer to them as rare species for simplicity). The number of occurrences per species ranged between 10 and 140, and species that were more threatened according to the national Red List of vascular plants in Switzerland (Moser *et al.* 2002) had lower numbers of occurrences (Fig. 2; see Table S1 for further details).

For this study, data on species occurrences (presence-only) were provided by the Swiss national data centre Info Flora (www.infoflora.ch;

Table 1. Environmental predictors used for modelling. A DTM with a resolution of 25 m was used for deriving the topographic predictors. The climatic predictors are means for the years 1961–1990

Predictor variables used for modelling		
Climatic predictors		
tave:	Mean annual temperature	Zimmermann & Kienast (1999)
prec:	Mean annual precipitation	Zimmermann & Kienast (1999)
pday:	Mean annual number of summer precipitation days	Zimmermann & Kienast (1999)
srad:	Annual global potential shortwave radiation	Algorithm following Kumar, Skidmore & Knowles (1997)
sfro:	Mean annual number of frost days during the growing season	Bolliger, Kienast & Zimmermann (2000)
Topographic predictors (based on a DTM with 25 m resolution)		
twi:	Wetness index	Following Tarboton 1997;
topo:	Topographic index	Camathias *et al.* 2013;
slp:	Slope	Camathias *et al.* 2013;
nness:	Northness	Camathias *et al.* 2013;
eness:	Eastness	Camathias *et al.* 2013;
Geological predictors		
caco3:	Calcareous content of the bedrock and surface material	Lehmann *et al.* (2010)

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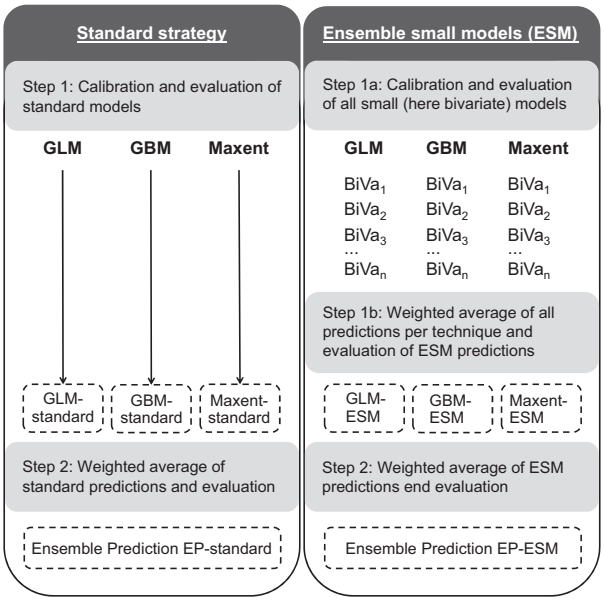

Fig. 1. Schematic of the two model strategies and the different steps of modelling. For the standard strategy (left panel) models of three standard techniques (GLM, GBM, Maxent) were calibrated and evaluated (step 1), and averaged to a single ensemble prediction (step 2). For the Ensemble Small Models (ESM) strategy (right panel) for each technique all bivariate models (BiVa; with 11 predictors $n = 55$) were calibrated and evaluated (step 1a), and averaged to a single ESM per technique (step 1b). ESMs were finally averaged again to a single ensemble prediction (step 2). Dashed lines indicate all models used for model comparisons.

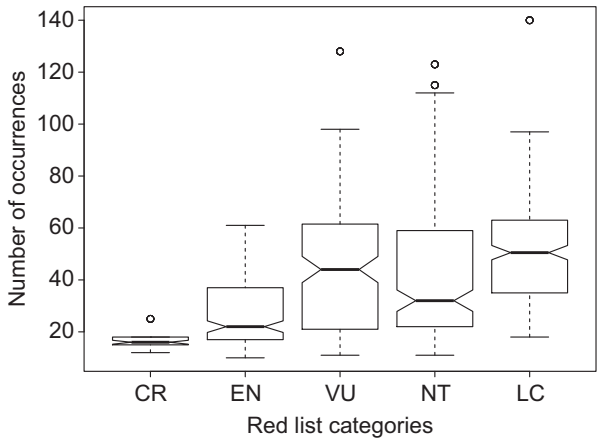

Fig. 2. Relationship between Red List status of the 107 target species (according to Moser *et al.* 2002) and the number of times they occurred. Species that were threatened (i.e., critically endangered (CR), endangered (EN) or vulnerable (VU)) or near threatened (NT) generally had a lower number of occurrences than non-threatened species (i.e., least concern (LC)).

see A1 in Appendix S1 for further information on the selection criteria of the target species). For analysis, groups of very rare species ($n \leq 25$ occurrences), rare species ($25 < n \leq 50$) and less rare species ($n > 50$) were built. The groups were well balanced, with 37 species in the very rare group, 34 in the intermediate group and 36 in the less rare group.

ENVIRONMENTAL PREDICTORS

From an initial set of 19 predictor layers with a 25-m resolution, 11 topo-climatic and geological predictors were selected by applying the rule of thumb that correlation coefficients of predictors from a set of 10 000 random background points should not exceed $|r| > 0{\cdot}7$ (Dormann *et al.* 2013). We kept the most direct predictors, for example temperature instead of altitude, and, due to the heterogeneous ecology of the species involved, we used the more general predictors, for example mean annual temperature instead of mean monthly temperature (Table 1).

MODEL CALIBRATION AND EVALUATION

All models were calibrated with presence-only data and 10 000 random background points that were selected from the entire area of Switzerland and kept constant for all species. Background points and occurrence points were weighted equally in the models (see A2 in Appendix S1 for a more detailed description of model implementation).

Fifty-fold split sampling (90% training data and 10% test data) was used to evaluate the models. In recent years, the modelling community has started a stimulating discussion about the most accurate way to evaluate model predictions, but a final consensus remains pending (for a review, see Franklin 2009). We thus used several indices to evaluate model performance such as the commonly used, but recently criticized, area under the receiver operating characteristic curve (AUC) (Lobo, Jiménez-Valverde & Real 2008; Peterson, Papeş & Soberón 2008; Jiménez-Valverde 2012; Smith 2013), the true skills statistic (TSS) (Allouche, Tsoar & Kadmon 2006), sensitivity, specificity and the recently developed but rarely used continuous Boyce index with a moving window (Hirzel *et al.* 2006; Petitpierre *et al.* 2012; see A3 in Appendix S1 for a more detailed description of the Boyce index and how it was implemented). The different indices for model evaluation were compared based on pairwise Pearson correlations.

We used an ensemble prediction (EP) by averaging outputs of the three standard SDM techniques based on Somers' D (i.e. rescaled AUC) weights to cope with model variability (Segurado & Araújo 2004; Elith & Graham 2009) and to improve reliability of the model predictions (Araújo & New 2007). Models with a Somers' D lower than 0 (i.e. AUC < 0·5 and hence worse than a random model) were set to zero and thus were not included in the EP. We used BIOMOD (Thuiller *et al.* 2009) to calibrate the models of the individual modelling techniques with the R package 'BIOMOD2' (version 2.1–40 and version 3.1–68 for the transferability assessment) but evaluated and averaged the models manually because ESMs are not implemented in BIOMOD2.

MODEL COMPARISONS

To test whether ESMs outperform standard SDMs, we applied linear mixed-effects models on the arcsine-transformed AUC and on the Fisher *z*-transformed Boyce index. We tested for effects of modelling strategies (ESMs, standard modelling approaches), modelling techniques (GLM, GBM, Maxent and their ensemble prediction EP), and sample size (i.e. categorical number of occurrences (n), with $n \leq 25$, $25 < n \leq 50$ and $n > 50$) and their two-way interactions (technique × strategy, technique × n, strategy × n). The effects of these variables were considered as fixed. The species were used as a random factor in these models. We used *P*-values for backward variable selection with the function LMER in the R package LME4.

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TRANSFERABILITY ASSESSMENT

We used a transferability assessment similar to Heikkinen, Marmion & Luoto (2012) to explore the predictive ability of ESMs to new areas. To do so, we split the occurrences of the 'intermediate rare' species group ($25 < n \leq 50$) into its eastern and western halves. The western halves of each species consist thus of ≤25 occurrences for each species and correspond therefore to our 'rare species' group. We used the western halves of the occurrences to calibrate the models, whereas the eastern halves were used for evaluation. We acknowledge that this kind of transferability assessment is not completely independent but likely the best solution given the rarity of the species. The modelling framework for the transferability assessment followed the framework described above with the exception that for model calibration the 10 000 pseudo-absences were only distributed in the western halves. Each model was first evaluated by the standard 'internal 50-fold split sampling' based on the calibration data set. To test for transferability, 10 000 pseudo-absences were extracted randomly from the eastern halves and were used with the eastern halves of the occurrences as an evaluation data set.

Results

Model evaluations based on AUC were highly correlated with evaluations based on TSS (which is in line with Allouche, Tsoar & Kadmon 2006), sensitivity and specificity (Pearson $|r| > 0{\cdot}7$ for all correlations; Fig. S2). In contrast, the correlations between the Boyce index and the other indices were much lower ($|r|$ ranging between 0·05 and 0·25; Fig. S2). Therefore, we present results only for AUC and the Boyce index in the following text (see Fig. S3 for results based on the remaining indices).

Out of 107 standard GLMs, the fitted probabilities of 36 models delivered values of zero or one. For these models, we did not calculate the Boyce index that needs at least three suitability values for calculation (see A3 in Appendix S1). Contrastingly, this was not an issue for ESM_{GLM}.

The mean AUC and mean Boyce index for the ESMs were 0·97 (±0·001 SE) and 0·88 (±0·005 SE), respectively. In comparison, the three standard SDM techniques and their ensemble prediction corresponded to a mean AUC of 0·93 (±0·003 SE) and a mean Boyce index of only 0·63 (±0·011 SE). Linear mixed-effects models on subsets of the data showed that ESMs nearly always outperformed their 'standard' counterparts (Table 2). The number of species performing better with ESMs compared to standard SDMs varied between 70 and 107 (Table 3). Differences between standard techniques and ESMs were particularly pronounced for GLMs for species with less than 50 occurrences when evaluated by AUC (Table 2 and Fig. 3). In the group consisting of the species that occurred most frequently ($n > 50$), however, ESMs were not superior to GBM or Maxent model performances when evaluated with AUC.

The effects of all fixed factors (i.e. effects of modelling technique, modelling strategy, number of occurrences and their two-way interactions) on AUC and the Boyce index were all highly significant (Table 4). For both evaluation measures, modelling strategy accounted for a large amount of deviance: the performance of ESMs was clearly better than that of standard SDMs, including their ensemble prediction (Fig. 3). The deviance accounted for by the fixed factor 'technique' was considerably lower than for 'strategy' (Table 4). However, strategy clearly interacted with sample size: The very rare species ($n \leq 25$) benefited most from the ESM strategy and the gain in model performance decreased with increasing sample size (Table 2 and Fig. 3). This benefit was much more pronounced for the Boyce index than for AUC. Nevertheless, the interaction between modelling strategy and sample size was significant for both evaluation measures ($P < 0{\cdot}001$; Table 4). For AUC in particular, the deviance of this interaction was high in comparison with the other terms fitted in the model. Red List status was correlated with sample size (Fig. 2), and thus, the most threatened species were also those with the greatest gain in model performance when ESMs were used.

Our results further showed that, for the standard strategies, the ensemble prediction performed better than the single modelling techniques (Fig. 3). For ESMs, however, ensemble predictions did not consistently perform better than ESMs with

Table 2. Differences in model accuracy between ESMs and standard SDMs

	ESMs vs. standard SDMs		
	*n*1	*n*2	*n*3
AUC			
EP	0·022 (±0·005)***	0·014 (±0·003)***	−0·002 (±0·002)*
GBM	0·035 (±0·009)***	0·016 (±0·003)***	0·001 (±0·002)
GLM	0·141 (±0·018)***	0·089 (±0·015)***	0·034 (±0·011)**
Maxent	0·029 (±0·008)***	0·015 (±0·004)***	0·000 (±0·003)
Boyce			
EP	0·245 (±0·025)***	0·136 (±0·014)***	0·086 (±0·01)***
GBM	0·284 (±0·017)***	0·259 (±0·016)***	0·252 (±0·021)***
GLM	0·417 (±0·053)***	0·309 (±0·038)***	0·229 (±0·028)***
Maxent	0·457 (±0·045)***	0·264 (±0·033)***	0·194 (±0·026)***

The table shows the average difference between ESMs and standard SDMs for AUC and the Boyce index, as well as standard errors (in brackets) and significance levels (***$P \leq 0{\cdot}001$, **$P \leq 0{\cdot}01$, *$P \leq 0{\cdot}05$). Linear mixed-effects models (type III Wald chi-square tests) were used to test for differences in model performance (arcsine-transformed AUC and Fisher z-transformed Boyce index) between standard SDMs and ESMs using subsets for each modelling technique (GBM, GLM, Maxent and ensemble prediction (EP)) and class of sample size ($n1 \leq 25$; $25 < n2 \leq 50$; $n3 > 50$). ESMs performed significantly better than standard SDMs, except for species with the highest number of occurrences for GBM and Maxent model when evaluation was based on AUC.

Table 3. Number of species (out of 107) which have higher AUC or Boyce scores with ESMs compared to standard SDMs (see also Table S1) for the three modelling techniques and their ensemble prediction

Technique	AUC	Boyce
EP	70	101
GBM	84	107
GLM	94	70
MAXENT	71	103

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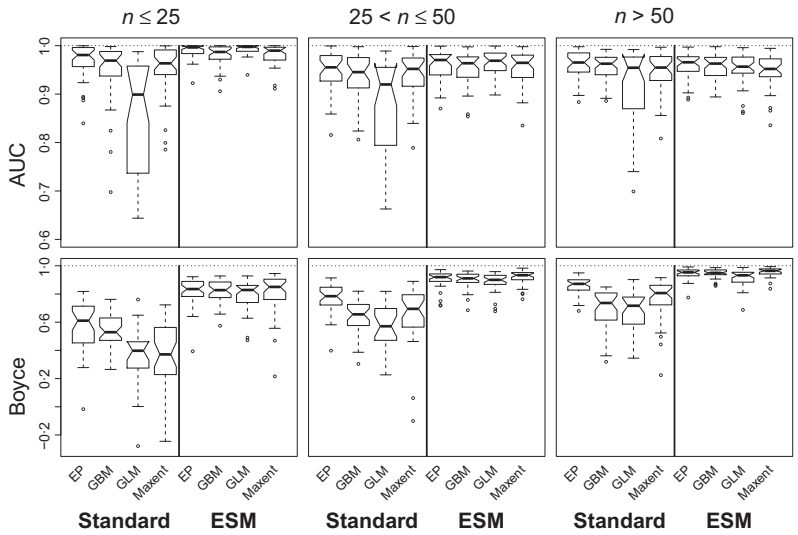

Fig. 3. Evaluation of the species distribution models using AUC (top) and the Boyce index (bottom). The number of species occurrences increases from the left to the right panels. Each panel is split into the two modelling strategies, 'Standard' and Ensemble Small Models 'ESM'. Each modelling strategy was applied on three modelling techniques (GLM, GBM and Maxent), which were averaged to build an ensemble prediction (EP). There are 37 species in the group $n \leq 25$, 34 species in the group $25 < n \leq 50$ and 36 species in the group $n > 50$. The model performance of a single species is the mean of 50 split sample runs.

Table 4. Analysis of deviance table (type III Wald chi-square tests) of a linear mixed-effects model showing that the performance of the models, evaluated by arcsine-transformed AUC and Fisher *z*-transformed Boyce index, depended significantly on the three fixed factors: class of sample size *n* ($n1 \leq 25$; $25 < n2 \leq 50$; $n3 > 50$), modelling technique (tech; GLM, GBM, Maxent and ensemble prediction) and modelling strategy (standard SDMs and ESMs). The interactions between the three fixed factors (modelling technique, modelling strategy and number of occurrences) were also significant. The 107 species were used as random effects

AUC	χ^2	d.f.	Pr(>χ^2)	Boyce	χ^2	d.f.	Pr(>χ^2)
(Intercept)	8911·6	1	<0·001	(Intercept)	688·2	1	<0·001
tech	16·2	3	0·001	tech	9·8	3	0·021
strategy	111·7	1	<0·001	strategy	166·2	1	<0·001
n	42·1	2	<0·001	n	146·7	2	<0·001
tech:n	24·2	6	<0·001	tech:n	43·1	6	<0·001
strategy:n	190·6	2	<0·001	strategy:n	20·5	2	<0·001
strategy:tech	276·6	3	<0·001	strategy:tech	69·2	3	<0·001

the single modelling techniques. Moreover, the intermodelling technique variation was much lower for ESMs than for the standard strategy, that is the predictions of the single modelling techniques were more similar when ESMs were used (Fig. 3 and Fig. S4).

TRANSFERABILITY ASSESSMENT

The evaluation of model performance of the transferability assessment is in line with the above shown evaluation of models based on the complete data set. The mean 50-fold split sampling AUC for ESMs based on the calibration data set was 0·95 (±0·004 SE) and the mean Boyce index was 0·82 (±0·007 SE) (Fig. S5). The evaluation scores of the standard SDMs were lower with a mean AUC of 0·92 (±0·007 SE) and a mean Boyce index of 0·62 (±0·02 SE).

Ensembles of small models were always outperforming standard SDMs when evaluated on the external evaluation data set (eastern occurrences; Fig. 4, see also Fig. S6). The mean AUC for ESMs on the geographically independent data set was 0·89 (±0·007 SE) and the mean Boyce index was 0·60 (±0·027 SE), whereas for the standard SDMs mean AUC was 0·80 (±0·012 SE) and the mean Boyce index was 0·14 (±0·04 SE). However, the performance between single modelling techniques was rather heterogeneous for ESMs when evaluation was based on Boyce index, with Maxent (AUC: 0·89 ± 0·013 SE; Boyce: 0·63 ± 0·048 SE), GLMs (AUC: 0·88 ± 0·014 SE; Boyce: 0·68 ± 0·045 SE) and EP (AUC: 0·89 ± 0·013 SE; Boyce: 0·61 ± 0·053 SE) performing better than GBMs (AUC: 0·88 ± 0·014 SE; Boyce: 0·50 ± 0·065 SE).

Discussion

Our results confirm that ESMs based on bivariate models are a powerful strategy for modelling rare species distributions. ESMs outperformed standard SDMs as well as their ensemble predictions, and the rarer the species was, the more pronounced the effect was. These results were found consistently with three modelling techniques.

ESMS AND ENSEMBLE MODELS

It is well known that model averaging provides more stabilized inference and better predictions (Burnham & Anderson 2002). Usage of ensembles in SDMs is however relatively new (Araújo & New 2007), yet the number of studies applying ensemble

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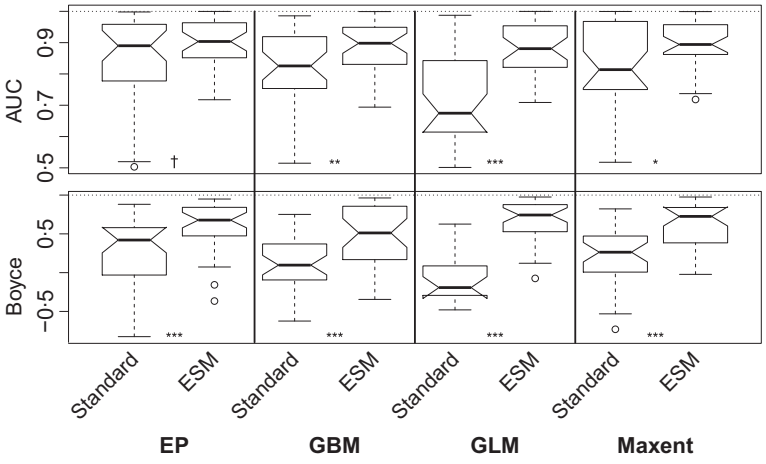

Fig. 4. Evaluation of the species distribution models following a transferability assessment using AUC (top) and the Boyce index (bottom). For the transferability assessment only the 34 species in the group $25 < n \leq 50$ (Fig. 3) were considered and split into one halve for calibrating and one halve for independently evaluating the models. Significance levels (***$P \leq 0.001$, **$P \leq 0.01$, *$P \leq 0.05$, $^{\dagger}P < 0.1$) between standard SDMs and ESMs are indicated for each technique (GBM, GLM and Maxent) and the ensemble prediction (EP).

SDMs has increased sharply within the last few years (e.g. Marmion *et al.* 2009; Coetzee *et al.* 2009; Lomba *et al.* 2010; Thuiller *et al.* 2011), and it has been shown that ensemble predictions perform better compared to single modelling techniques (Marmion *et al.* 2009).

Our results are consistent with this finding in that averaged models perform better than models based on single modelling techniques: the ensemble predictions of our standard SDMs performed better than the single modelling techniques (Fig. 3). Averaging ESMs across model techniques (ESM_{EP}) however did not further increase model performance (Fig. 3) and there is thus no need to build ESM_{EP} which are more complex and more computationally intensive than ESMs based on a single modelling technique.

We found that ESM quickly come close to their final performance (e.g. in terms of AUC), that is model performance increases asymptotically (Fig. S7). Interestingly, there were nearly always some combinations of a small subset of bivariate models that outperformed the full ESM. It would thus be possible to further optimize the ESM framework by searching for best subsets of all combinations of bivariate models.

Standard SDMs performed poor when the evaluation was based on the transferability assessment while ESMs were performing much better (Fig. 4). Single ESMs based on Maxent, GLM and EP were also better performing than ESMs based on GBMs, at least when they were evaluated by Boyce index. As a consequence, we recommend the use of ESM_{Maxent} or ESM_{GLM} for modelling rare or under-sampled species because they perform well in both internal (i.e. the whole data set Fig. 3, see also Fig. S5) and external cross-validation (i.e. the evaluation data set of the transferability assessment Fig. 4, see also Fig. S6).

Ensembles of small models are not restricted to bivariate models. Future investigations should also consider uni- or trivariate (or higher) models to build ESMs.

MODELLING RARE SPECIES

Although it is well known that modelling rare species is often problematic in terms of model overfitting and inaccurate model predictions (Hernandez *et al.* 2006; Barry & Elith 2006; Wisz *et al.* 2008), few studies have paid sufficient attention to this problem. Maxent has typically been chosen for modelling rare species (Parolo, Rossi & Ferrarini 2008; Gogol-Prokurat 2011; Niamir *et al.* 2011; Bean, Stafford & Brashares 2012) because this technique performs comparatively well with low sample sizes (Hernandez *et al.* 2006; Wisz *et al.* 2008). In our study, however, standard Maxent models did not perform better with very rare species than other standard techniques. Moreover, Maxent ESMs clearly outperformed standard Maxent SDMs (Fig. 3). The transferability assessment complementarily showed that the best predictions are obtained when ESMs are implemented with Maxent and GLMs, two approaches closely related to each other and to traditional statistics (hence a form of 'back-to-basics'). GLMs in particular have already been shown to be superior to GAM when projecting SDMs to a different area (Randin *et al.* 2006).

The problems associated with modelling rare species are often not fully recognized because common evaluation indices based on presence–absence information, like AUC, generally give higher scores than presence-only indices, like Boyce, when the number of presences is low (i.e. the justification for the Boyce index; Hirzel *et al.* 2006). The AUC index, one of the most widely used indices, is known to overestimate the performance of models calibrated for rare species (Lobo, Jiménez-Valverde & Real 2008; Peterson, Papeş & Soberón 2008; Smith 2013).

Higher AUC values of rare species compared to more common ones, which were also found with the standard strategy in our study (Fig. 3), imply excellent model predictions (AUC > 0·9 according to Araújo *et al.* (2005)). However, the trend towards high AUC values for models of rare species does not hold when the models are evaluated on independent data (Fig. 4 and Fig. S6). Hence, the limitation of modelling rare species could remain unnoticed in many studies without a transferability assessment. Contrary to AUC, rare species perform worse compared to the less rare ones when evaluated with the Boyce index (Fig. 3), and we advocate, in line with other studies (Hernandez *et al.* 2006), that researchers should not rely solely on the AUC if no true absences are available. Instead, we suggest using the Boyce index more systematically, and choosing ESMs when modelling rare species.

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LIMITATIONS

Our study is not without limitations. One first shortcoming of the ESM approach is its higher computational effort in comparison with standard SDM approaches. The current lack of implementation capability in statistical software, such as R, has also limited its use so far, but we now provide R scripts for building ensembles of bivariate models in R (see www.purl.org/wsl/esm).

Another limitation is that we randomly selected 10 000 background points located throughout the entire area of Switzerland for calibrating the models, and we kept this background constant for all species. Our 107 target species were rare and many were restricted to small regions. As a consequence, for some species many background points could be located far away from presence points, leading to high AUC values by easily distinguishing the background from presence points (VanDerWal *et al*. 2009). Other strategies to select the background data and the usage of true absence or abundance data could thus be tested in future assessments of the ESM approach.

A third limitation is related to the options used to parameterize the standard SDMs, and especially variable selection. Our implementation of the ESM approach may be seen as incorporating a type of variable selection, through the removal of too poor bivariate models (AUC $\leq$ 0·5) and the weighting of the remaining bivariate models by their performance. However, variables were never completely eliminated from ESMs (Fig. S1). ESMs therefore did not reduce the set of variables, like variable selection based on stepwise regression or using an all subset approach (which was also tested and confirmed the superior performance of ESMs; see Figs S8 and S9). ESMs, therefore, rather keep each variable but weight them according to the evaluation of the single small models, comparable to multi-model inference. Even models based on low sample sizes include therefore the whole environmental information which seems to be – besides the reduction of model overfitting – a main reason for the better performance of ESMs.

A last limitation of our study is that we did not assess the effect of spatial autocorrelation. A common problem of SDMs is that spatial autocorrelation may bias parameter coefficients (Dormann *et al*. 2007), which may in turn lead to low transferability of the predictions to new areas (Segurado, Araújo & Kunin 2006; Heikkinen, Marmion & Luoto 2012; Wenger & Olden 2012). Cross-validation methods that randomly select subsets of test data for model evaluation might therefore suggest accurate models in the presence of spatial autocorrelation, but transferability to other regions (or time periods) would then be reduced. The transferability assessments we performed in this study suggest that if some level of spatial autocorrelation was present in our data, it did not affect the better performance of ESM compared to standard ensemble SDMs and therefore our main conclusions. Due to the characteristic of rare species with a restricted distribution, the most adequate way to implement a transferability assessment was in our case by dividing the data set into two halves, but other types of transferability assessments could be tested in subsequent ESM studies.

Conclusions

Ensembles of small models represent a powerful strategy applicable with various standard SDM techniques. The strength of ESMs is their high performance with small sample sizes, a characteristic of data for rare and threatened species, or for elusive species that are hard to detect. The potential of ESMs was further demonstrated by their ability to predict to independent areas. Here, we only tested ESMs with vascular plants, but ESMs are expected to perform as well with other taxonomic groups. We encourage future studies to use ESMs when rare species distributions need to be modelled because of their outstanding performance. In addition to the introduction of modern machine-learning techniques, like Maxent and GBMs, and the application of averaging single SDM techniques into ensemble predictions, implementing ESMs and their further optimization (e.g. allowing inclusion of less or more than two predictors at a time) should be considered as a next step for improving species distribution models for rare species.

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Data accessibility

Information about the species used in this study can be found in the supporting information (Table S1). Data on species occurrences (presence-only) were provided by the Swiss national data centre Info Flora (www.infoflora.ch) and are archived there. An R script to compute ESMs is uploaded as online supporting information (www.purl.org/wsl/esm).

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

Fig. S1. Number of bivariate models used for ESMs and variable importance for ESMs and standard SDMs.

Fig. S2. Correlation pairs and histograms of model performance from six evaluation indices: AUC, TSS, sensitivity, specificity, Boyce index and MPA.

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Fig. S3. Model performance according to TSS, sensitivity (Se), specificity (Sp) and (MPA) and its relationship with different modelling strategies and techniques.

Fig. S4. Boxplots of Pearson correlation coefficients between predictions of single modelling techniques.

Fig. S5. Internal evaluation of the transferability assessment.

Fig. S6. Internal vs. external evaluation of the transferability assessment.

Fig. S7. AUC against an increasing number of averaged bivariate models.

Fig. S8. Boxplots of the performance of ESMs in comparison to different approaches to build standard SDMs using internal evaluation cross-validation.

Fig. S9. Boxplots of the performance of ESMs in comparison to different approaches to build standard SDMs using external validation of the transferability assessment.

Table S1. Complete species list with species specific information.

Appendix S1. Details on the Methods.

Appendix S2. Commented R code for ESMs (available on www.purl.org/wsl/esm).

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